National University of Computer and Emerging Sciences,
Lahore Campus

	Course Name:	Computer Networks	Course Code:	CS 3001
	Program:	BS (Software Engineering)	Semester:	Spring 2026
	Duration:	15 minutes	Total Marks:	15
	Paper Date:	23-April-2026	Section	6A , 6B
	Exam Type:	Quiz 5 - Chapter 5	Page(s):	2

Student Name **Roll No.** **Section:**

Q1. Encircle the correct option. **[5 Marks] [CLO 3]**

1. **BGP is a _______________ protocol.**
   a. Distance Vector
   b. Link State Protocol
   c. Path Vector Protocol
   d. None of the above

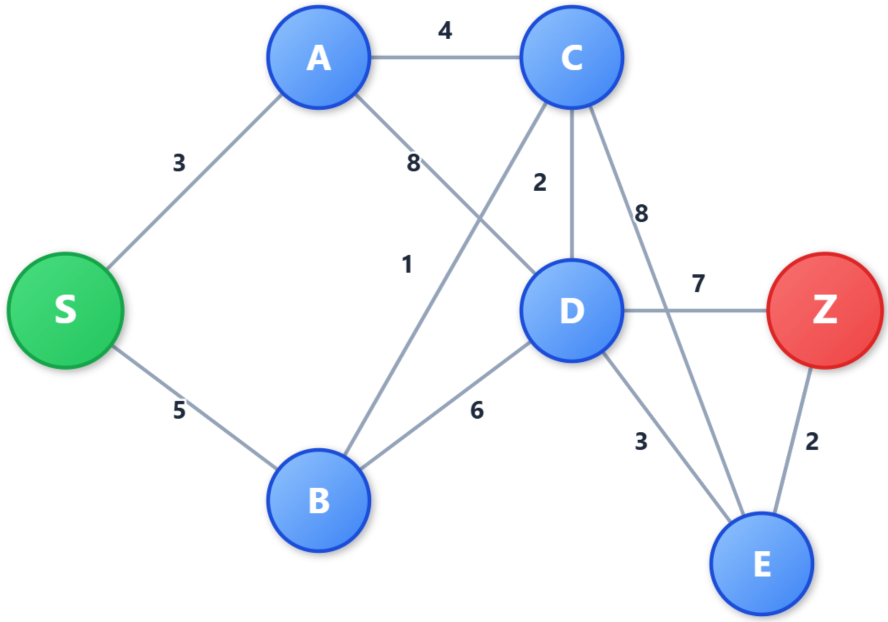
2. **In the context of Link-State routing algorithms, which of the following scenarios is the primary cause of 'routing oscillation'?**
   a. The use of link costs that are dynamically calculated based on traffic volume (load.)
   b. A topology having multiple equal path costs to the same destination
   c. A delay in the flooding of Link State Advertisements (LSAs) across the network
   d. Routing oscillations do not occur in link state algorithms

3. **When a router in an Autonomous System (AS) has multiple possible exit points to reach a destination in a different AS, which criterion does it use to implement 'hot potato' routing?**
   a. It randomly alternates between all available paths to all exit points to ensure load balancing
   b. It selects the path to the exit point with the lowest internal cost (lowest intra-AS cost)
   c. It selects the shortest path to the final destination
   d. It opts for cold potato instead of hot potato routing

4. **A router learns a route to 10.0.0.0/8 via BGP with Next-Hop 192.0.2.1. However, the router's routing table does not have an entry for 192.0.2.1. What is the result?**
   a. The router installs the route but marks it as "unreachable."
   b. The router initiates an ARP request for 192.0.2.1.
   c. The BGP route is considered invalid and is not installed in the forwarding table.
   d. The router uses the default gateway to forward traffic for this destination.

5. **Which property is TRUE about Distance Vector (DV) algorithms?**
   a. Require full topology knowledge
   b. Use centralized computation
   c. Converge via iterative neighbor exchange
   d. Never suffer from routing loops

<u>Q2.</u> Consider the network shown below and assume that each node initially knows the costs to each of its neighbors. Consider the link state algorithm and compute shortest path from S to all other nodes by filling the below table step by step. **[10 Marks] [CLO 3]**



Step	Set N'	D(A), p(A)	D(B), p(B)	D(C), p(C)	D(D), p(D)	D(E), p(E)	D(Z), p(Z)
0							
1							
2							
3							
4							
5							
6							
7							
8							
9							
10							